

Clifford Space Networks: Sub-Quadratic Non-Abelian Lie Rotor Scans for Geometric Sequence Modeling and Physical Systems

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Abstract

Modern deep learning architectures predominantly map representations to unconstrained Euclidean coordinates in $\mathbb{R}^d$, neglecting coordinate-free geometric invariants and non-commutative compositionality. While Transformer self-attention achieves high expressive capacity, it incurs an irreducible quadratic computational complexity $\mathcal{O}(L^2)$ and an exploding KV-cache memory footprint. Conversely, sub-quadratic State Space Models (SSMs) enforce strictly commutative diagonal transitions ($A_t A_{t-1} = A_{t-1} A_t$), leading to state degeneracy when capturing hierarchical order permutations and non-commutative dependencies.

This work introduces **Clifford Space Networks (CSN)**, an analytical geometric framework founded on Clifford multivector algebras $\mathcal{Cl}(p, q)$ and continuous **non-abelian** Lie rotor groups $\text{Spin}(n)$. Unlike existing Clifford neural networks designed primarily for static PDE surrogates and spatial equivariance, CSN formulates recurrent state transitions as isometric Lie group rotations via the rational Cayley transform, executing in strictly sub-quadratic time through an $\mathcal{O}(L \log L)$ parallel associative scan and $\mathcal{O}(1)$ autoregressive generation memory.

As a v1 feasibility study, this paper validates the core mathematical operator across compact sequence and spatial perception tasks:

1. **Sequence Feasibility (CSN-LM)**: On the Tiny Shakespeare benchmark, CSN-LM $\mathcal{Cl}(8, 0)$ reaches **50.50% Top-1 Accuracy** (PPL 5.46) with only **158,164 parameters**, matching the accuracy of a 1.90M parameter Scaled Transformer (**19.7× parameter reduction**) and demonstrating up to **1.54× faster test-time token generation** at $\mathcal{O}(1)$ state memory (~ 21 MB). While these compact results confirm exceptional sample efficiency, large-scale multi-billion-token benchmarks (BabyLM 10M/100M, TinyStories) are currently underway to evaluate regimes where attention mechanisms traditionally scale.
2. **Spatial Feasibility (CSN-V3)**: On MOT16 pedestrian detection, CSN-V3 achieves **70.0% recall** and **50.19 F1 per million parameters** at 1.29M parameters (1.98× fewer than YOLO26n) with native fused CUDA scan kernels operating at real-time speeds (54.98 ms / 18.2 FPS on a consumer GTX 1660 Ti).

Finally, we position CSN primarily as an **analytical solver for formal logic graphs and physical Hamiltonian systems**, diagnose the **interconnect bottleneck** of information-dense Clifford neurons, and outline a principled architectural evolution toward **Port-Hamiltonian Interconnections (PHS)** and full-scale ImageNet benchmarks.

1 Introduction

The modeling of dynamic temporal processes, non-commutative symbolic structures, and continuous physical systems remains constrained by foundational algebraic trade-offs across modern neural architectures.

1.1 The Non-Abelian Dilemma in Sequence Modeling

In sequence modeling and natural language processing, the Transformer architecture [10] relies on unconstrained pairwise dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Although empirically dominant at scale, self-attention incurs an irreducible quadratic computational bottleneck $\mathcal{O}(L^2)$ and an exploding Key-Value (KV) cache memory footprint during autoregressive generation.

To achieve linear computation, State Space Models (SSMs) such as S4, S5, and Mamba [3] introduce input-dependent linear recurrences. However, to enable the fast parallel prefix scan (Blelloch scan) in $\mathcal{O}(L \log L)$, SSMs enforce that the state transition matrix A_t is strictly **diagonal**:

$$h_t = A_t h_{t-1} + B_t x_t, \quad A_t = \text{diag}(a_{t,1}, \dots, a_{t,N}) \quad (2)$$

Consequently, consecutive state transitions are strictly **commutative** ($A_t A_{t-1} = A_{t-1} A_t$). In contrast, grammatical syntax, algorithmic reasoning, and formal logic are fundamentally **non-commutative**; the semantic state depends critically on token order and non-trivial permutations ($A \circ B \neq B \circ A$). To capture non-commutative hierarchies, commutative SSMs must significantly inflate their hidden state dimensions.

Similarly, Closed-form Continuous-time (CfC) networks [4] model dynamical systems via continuous ODEs with explicit decay:

$$h_t = \sigma(-f(x_t)\Delta t) \odot \tanh(W_1 x_t) + (1 - \sigma(-f(x_t)\Delta t)) \odot \tanh(W_2 x_t) \quad (3)$$

While continuous, CfC couples inputs and recurrent states inside non-linear saturating activations ($\tanh$), which precludes formulation as an associative matrix scan and restricts training to sequential $\mathcal{O}(L)$ loops.

1.2 Distinction from Prior Clifford Neural Networks

The integration of Clifford (geometric) algebras into machine learning has garnered significant attention, particularly through Clifford Neural Layers for PDE modeling [1], Geometric Clifford Algebra Networks (GCANs) [7], Clifford Group Equivariant Networks (CGENN) [6], and Geometric Algebra Transformers (GATr) [2].

However, a fundamental architectural distinction separates CSN from prior literature:

- **Target Domain:** Prior Clifford networks focus almost exclusively on static spatial symmetry groups ($O(n), E(n)$) or surrogate modelling for continuous Partial Differential Equations (e.g., fluid dynamics, weather forecasting). In contrast, CSN targets **sub-quadratic dynamic sequence modeling and state space transitions**.
- **Transition Mechanics:** While previous works apply static geometric products of multivector weights ($W \star x$), CSN formulates recurrent state updates via the **rational Cayley transform** on Lie algebras $\mathfrak{so}(n) \rightarrow \text{Spin}(n)$.
- **Associative Scans:** Prior multivector architectures do not formulate causal temporal recurrences or require $\mathcal{O}(L^2)$ attention (as in GATr). CSN derives an exact affine closed form that admits a **parallel prefix scan in $\mathcal{O}(L \log L)$** and registers-level fused GPU kernels.

1.3 Primary Orientation: Analytical Graph and Physical Solvers

While standard deep learning architectures approximate tasks through statistical pattern matching on flat Euclidean tensors, Clifford Space Networks were conceived as an **analytical engine**:

1. **Analytical Logic Graph Solvers:** In formal logic, causal dependency networks, and knowledge graphs, relations are non-commutative and structured. By mapping entities to multivectors and relations to Spinor Lie rotors $R_{ij} \in \text{Spin}(n)$, logical inference along graph paths reduces to exact algebraic rotor composition without geometric distortion.
2. **Physical System Modeling and Gauge Invariants:** Physical conservation laws (Hamiltonian mechanics, Maxwell’s electrodynamics $F = E + IcB$, and gauge field theories) are natively expressed via differential forms and bivectors. CSN operates directly on the Lie group $\text{Spin}(n)$ with strict algebraic norm preservation, providing an inductive bias for physical dynamical systems.

1.4 Architectural Scope: v1 Feasibility and the Road to Scale

This work constitutes a **Version 1 (v1) architectural feasibility study**. The empirical benchmarks presented herein validate that non-abelian Lie rotor scans can be executed with exact isometry, sub-quadratic training, and constant inference memory.

We explicitly acknowledge that while CSN demonstrates remarkable parameter efficiency on compact benchmarks, standard Transformers possess established scaling properties that emerge at multi-billion-token scales. Systematic evaluations on standardized corpora (BabyLM 10M/100M, TinyStories) and scaling spatial vision to ImageNet-1k are under active execution to characterize asymptotic scaling laws.

2 Mathematical Foundations: Clifford Algebras and Lie Rotors

2.1 Clifford Algebra $\mathcal{Cl}(p, q)$

Let $V = \mathbb{R}^{p+q}$ be an n -dimensional real vector space equipped with a quadratic form Q of signature (p, q) where $n = p + q$. The Clifford algebra $\mathcal{Cl}(p, q)$ is generated by V subject to:

$$v^2 = Q(v) \cdot 1, \quad \forall v \in V \quad (4)$$

Given an orthonormal basis $\{e_1, \dots, e_n\}$, the generators satisfy:

$$e_i e_j + e_j e_i = 2\eta_{ij} \mathbf{1}, \quad \eta_{ij} = \text{diag}(\underbrace{+1, \dots, +1}_p, \underbrace{-1, \dots, -1}_q) \quad (5)$$

A multivector $A \in \mathcal{Cl}(p, q)$ admits a canonical grade decomposition:

$$A = \langle A \rangle_0 + \langle A \rangle_1 + \langle A \rangle_2 + \dots + \langle A \rangle_n = \sum_{k=0}^n \langle A \rangle_k \quad (6)$$

where $\langle A \rangle_k \in \bigwedge^k V$ corresponds to scalars ($k = 0$), vectors ($k = 1$), bivectors ($k = 2$), and pseudo-scalars ($k = n$).

2.2 Spinor Lie Group and the Rational Cayley Transform

The even sub-algebra $\mathcal{Cl}^+(p, q) = \bigoplus_{k \text{ even}} \bigwedge^k V$ defines the Spinor group:

$$\text{Spin}(p, q) = \{R \in \mathcal{Cl}^+(p, q) \mid R\tilde{R} = 1, \forall v \in V : Rv\tilde{R} \in V\} \quad (7)$$

where $\tilde{R}$ is the reversion anti-automorphism. In the Euclidean signature $\mathcal{Cl}(n, 0)$, $\text{Spin}(n)$ doubly covers $\text{SO}(n)$:

$$1 \rightarrow \mathbb{Z}_2 \rightarrow \text{Spin}(n) \xrightarrow{\rho} \text{SO}(n) \rightarrow 1, \quad \rho(R)(v) = Rv\tilde{R} \quad (8)$$

The Lie algebra $\mathfrak{spin}(n) \cong \mathfrak{so}(n)$ consists of pure bivectors $\langle A \rangle_2 = \sum_{i < j} b_{ij} e_i \wedge e_j$.

To map bivectors $\Omega \in \mathfrak{so}(n)$ into group rotations without numerical truncation, CSN applies the **rational Cayley transform**:

$$R = \mathcal{C}(\Omega) = (I + \Omega)(I - \Omega)^{-1} \in \text{Spin}(n) \quad (9)$$

The Cayley transform is strictly isometric:

$$\|Rv\| = \|v\|, \quad \forall v \in \mathbb{R}^n \quad (10)$$

guaranteeing bounded spectral radius and zero gradient vanishing/explosion by geometric construction.

2.3 Exact Clifford Closed Form vs. Heuristic CfC

In contrast to empirical bounding functions on flat vectors [4], CSN defines an **exact algebraic closed form**:

$$\Omega_t = \frac{1}{2}B(x_t)\Delta t_t, \quad R_t = (I + \Omega_t)(I - \Omega_t)^{-1} \in \text{Spin}(n) \quad (11)$$

$$h_t = \sqrt{\alpha_t}R_th_{t-1} + (1 - \alpha_t)S_t \equiv M_th_{t-1} + C_t \quad (12)$$

where $\alpha_t = \sigma(-\gamma(x_t)\Delta t_t) \in (0, 1)$ is continuous decay, $M_t = \sqrt{\alpha_t}R_t$, and $C_t = (1 - \alpha_t)S_t$.

Because $(M_j, C_j) \circ (M_i, C_i) = (M_jM_i, M_jC_i + C_j)$ satisfies exact mathematical associativity:

$$((M_k, C_k) \circ (M_j, C_j)) \circ (M_i, C_i) = (M_k, C_k) \circ ((M_j, C_j) \circ (M_i, C_i)) \quad (13)$$

the entire sequence can be trained via the **parallel associative prefix scan** in $\mathcal{O}(L \log L)$ time.

3 Asymptotic Complexity Analysis

Table 1 compares the asymptotic characteristics of modern architectures.

Table 1: Asymptotic complexity and architectural characteristics across foundation paradigms.

Architecture	Training Time	Inference Latency	Inference Memory	State Algebra	Parallel Scan
Transformer (GPT / ViT)	$\mathcal{O}(L^2d + Ld^2)$	$\mathcal{O}(Ld)$	$\mathcal{O}(Ld)$	Flat Dot-Product	Exact ($\mathcal{O}(1)$ Matrix)
Mamba S6 (Gu et al.)	$\mathcal{O}(Ldd_{\text{state}})$	$\mathcal{O}(dd_{\text{state}})$	$\mathcal{O}(d_{\text{state}})$	Commutative Diagonal	Associative Scan
Liquid CfC (Hasani et al.)	$\mathcal{O}(Ld)$	$\mathcal{O}(d^2)$	$\mathcal{O}(d)$	Euclidean Vector ODE	None (Sequential Loop)
CSN-LM (Parallel Scan)	$\mathcal{O}(L \log L \cdot \text{ND}^2)$	$\mathcal{O}(\text{ND}^2)$	$\mathcal{O}(\text{ND}^2)$	Non-Abelian Spin(n)	Associative Prefix Scan
CSN-LM (Fused CUDA)	$\mathcal{O}(L \cdot \text{ND}^2)$	$\mathcal{O}(\text{ND}^2)$	$\mathcal{O}(\text{ND}^2)$	Non-Abelian Spin(n)	Single-Launch Kernel

4 Feasibility Benchmark: Dynamic Sequence Modeling

4.1 Architecture of CSN-LM

In CSN-LM, input token sequences $x \in \{1, \dots, V\}^L$ are projected into embedding dimension $\mathbb{R}^d$. Each Clifford recurrent layer computes:

- Input State Injection:** $S_t = W_{\text{in}}x_t \in \mathbb{R}^{N \times D \times D}$ for N Clifford neurons and dimension D .
- Continuous Adaptive Pacing:** $\Delta t_t = \text{softplus}(W_{\Delta t}x_t) \cdot \Delta t_{\text{base}}$.

3. **Continuous Conductance Decay:** $\alpha_t = \sigma(-\text{softplus}(W_\gamma x_t)\Delta t_t) \in (0, 1)$.
4. **Non-Abelian Generator:** $\Omega_t = \frac{1}{2}\Delta t_t \sum_{i < j} b_{ij}(x_t)(E_{ij} - E_{ji}) \in \mathfrak{so}(D)$.
5. **Rational Cayley Rotor:** $R_t = (I + \Omega_t)(I - \Omega_t)^{-1} \in \text{Spin}(D)$.
6. **Affine Recurrence:** $h_t = \sqrt{\alpha_t}R_th_{t-1} + (1 - \alpha_t)S_t$.

To decode logits without unconstrained Euclidean projections, CSN-LM employs a **Hyperbolic Cosine Centroid Head**:

$$\text{logit}_c = s \cdot \cos(d_g(h_t, \mathcal{E}_c)) + b_c = s \cdot \frac{\langle h_t, \mathcal{E}_c \rangle}{\|h_t\| \|\mathcal{E}_c\|} + b_c \quad (14)$$

4.2 Sequence Modeling Feasibility Results

Table 2 evaluates CSN-LM against sequence baselines on the Tiny Shakespeare benchmark (1.1M characters).

Table 2: Language modeling performance on the Tiny Shakespeare feasibility benchmark.

Model Architecture	State Dim / Neuron	Parameters	PPL	Top-1 Acc (%)	Bits/Char
Transformer (2 layers, 4 heads)	N/A	240,000	11.75	28.42%	3.554
Mamba S6 (2 layers, $d_{\text{state}} = 16$)	16 (diagonal)	72,000	7.37	41.65%	2.882
Liquid CfC (2 layers stacked)	64 (Euclidean)	86,400	9.12	36.80%	3.189
CSN-LM $\mathcal{Cl}(4, 0)$	16 (SO(4) rotors)	80,802	6.36	46.23%	2.669
CSN-LM $\mathcal{Cl}(6, 0)$	64 (SO(8) rotors)	126,740	5.68	49.07%	2.506
CSN-LM $\mathcal{Cl}(8, 0)$ (Master)	256 (SO(16) rotors)	158,164	5.46	50.50%	2.449

4.3 Iso-Accuracy Scaling Parity

To reach the accuracy of CSN-LM with only 158k parameters, a standard Transformer was scaled until matching parity ($\approx 50.0\%$ Top-1 Accuracy, Table 3).

Table 3: Iso-accuracy scaling benchmark: parameters required to reach $\approx 50\%$ Top-1 Accuracy.

Model Architecture	Parameters	Multiplier	PPL	Top-1 Acc (%)
CSN-LM $\mathcal{Cl}(8, 0)$	158,164	1.0$\times$	5.46	50.50%
Scaled Causal Transformer (4L, $d = 192$)	1,900,000	19.7$\times$	5.51	49.80%
Scaled Mamba S6 (2L, $d = 128$, $d_{\text{state}} = 16$)	460,000	2.9 $\times$	6.42	44.10%
Scaled Liquid CfC (2L, $d = 128$)	380,000	2.4 $\times$	7.89	40.20%

The Transformer requires 1.90 million parameters (19.7 $\times$ overhead) to match CSN-LM on this benchmark, highlighting the strong inductive bias of non-abelian rotor memory under strict capacity constraints.

5 Feasibility Benchmark: Static Spatial Perception

To evaluate geometric rotors on static 2D vision, CSN-V3 employs dual-grade streams $\mathcal{F}_{\text{spatial}} = \mathcal{Cl}(4, 0) \parallel \mathcal{Cl}(8, 0)$ regularized by Sasaki tangent bundle metrics [8]:

$$g_S(X, Y) = g(\pi_* X, \pi_* Y) + g(\mathcal{K}X, \mathcal{K}Y) \quad (15)$$

Table 4 reports pedestrian detection performance on MOT16 at 640×640 resolution on an NVIDIA GeForce GTX 1660 Ti with native fused CUDA kernels enabled.

Table 4: Computer Vision benchmark and execution metrics on NVIDIA GeForce GTX 1660 Ti (6 GB VRAM) at 640×640 resolution.

Model Architecture	Parameters	Pre-training Dataset	Latency	FPS	Peak VRAM	Recall (%)	F1 (%)	Info Density
YOLO26n (Calibrated Baseline) [†]	2,572,280	MS COCO (118k img, 500 ep)	46.78 ms	21.4	80.7 MB	65.4%	73.60%	29.39
CSN-V1 (Surgical Baseline)	950,510	MOT16 Scratch (5.3k fr, 15 ep)	49.27 ms	20.3	107.6 MB	60.4%	59.95%	64.65
CSN-V3 Deep Unleash (Peak)	1,296,213	MOT16 Scratch (5.3k fr, 20 ep)	54.98 ms	18.2	107.6 MB	61.0%	65.06%	50.19
CSN-V3 High-Sensitivity ($\tau = 0.15$)	1,296,213	MOT16 Scratch (5.3k fr, 20 ep)	56.79 ms	17.6	107.6 MB	70.0%	60.31%	46.53

Despite training exclusively on MOT16 (5,316 frames) compared to YOLO26n’s full MS COCO pretraining (118,287 images), CSN-V3 achieves **70.0% recall** and **50.19 F1/M-Param** (+70.7% information density) at $1.98\times$ fewer parameters, with register-level fused CUDA kernels delivering real-time throughput (54.98 ms / 18.2 FPS).

6 Real-Hardware Autoregressive Generation Metrics

Table 5 reports measured autoregressive generation throughput on GTX 1660 Ti across context horizons $L \in [128, 2048]$.

Table 5: Real-hardware autoregressive test-time generation benchmark on NVIDIA GeForce GTX 1660 Ti (6 GB VRAM).

Architecture	Params	Acc Parity	$L = 128$	$L = 512$	$L = 1024$	$L = 2048$	State Memory
Scaled Transformer (4L, $d = 192$)	1,900,000	49.80% (Parity)	201.3 tok/s	244.6 tok/s	195.8 tok/s	195.6 tok/s	$\mathcal{O}(L)$ (KV-Cache)
Standard Transformer (2L, $d = 64$)	240,000	28.42% (Sub-Par)	552.4 tok/s	417.1 tok/s	459.3 tok/s	402.4 tok/s	$\mathcal{O}(L)$ (KV-Cache)
Mamba S6 (2L, $d_{\text{state}} = 16$)	72,000	41.65% (Sub-Par)	103.3 tok/s	98.4 tok/s	88.7 tok/s	81.2 tok/s	$\mathcal{O}(1)$ (Constant)
Liquid CFC (2L, $h = 64$)	86,400	36.80% (Sub-Par)	39.0 tok/s	36.2 tok/s	31.5 tok/s	27.8 tok/s	$\mathcal{O}(1)$ (Constant)
CSN-LM $\mathcal{C}\ell(4, 0)$	80,802	46.23%	282.5 tok/s	232.1 tok/s	257.9 tok/s	271.8 tok/s	$\mathcal{O}(1)$ (20.7 MB)
CSN-LM $\mathcal{C}\ell(6, 0)$	126,740	49.07%	273.9 tok/s	252.0 tok/s	259.8 tok/s	301.0 tok/s	$\mathcal{O}(1)$ (21.1 MB)
CSN-LM $\mathcal{C}\ell(8, 0)$ (Master)	158,164	50.50% (Parity)	259.7 tok/s	224.1 tok/s	231.5 tok/s	259.0 tok/s	$\mathcal{O}(1)$ (21.6 MB)

At iso-accuracy parity, CSN-LM $\mathcal{C}\ell(6, 0)$ generates at **301.0 tokens/second** at $L = 2048$ ($1.54\times$ speedup over the Scaled Transformer at 195.6 tok/s), while maintaining a strictly flat $\mathcal{O}(1)$ state memory footprint (~ 21 MB).

7 The Interconnect Bottleneck and Future Directions: Port-Hamiltonian Interconnections (PHS)

7.1 Diagnosing the Interconnect Bottleneck in v1

A primary analytical insight emerging from the v1 implementation of Clifford Space Networks concerns the disparity between internal neuronal capacity and inter-layer communication:

Information-Dense Neurons vs. Euclidean Linear Pipes: Each Clifford neuron operates as a high-dimensional geometric entity containing a full multivector sub-algebra with up to 256 degrees of freedom. However, in the current v1 architecture, inter-neuron feature mixing is mediated by standard Euclidean linear projection weights W .

This mismatch creates an **interconnect bottleneck**: dense, geometrically structured multivector representations are forced through flat Euclidean projections, disrupting geometric coherence and inducing representation bottlenecks across deep stacks.

7.2 Toward Port-Hamiltonian Systems (PHS) for Inter-Neuron Coupling

To resolve this interconnect deficit without resorting to ad-hoc projection matrices, future iterations of CSN will formulate the inter-neuron connection graph as a **Port-Hamiltonian**

System (PHS) [9].

In port-Hamiltonian theory, dynamic energy exchange between interconnected subsystems is governed by skew-symmetric Dirac structures and power-conserving ports:

$$\begin{pmatrix} \dot{x} \\ y \end{pmatrix} = \begin{pmatrix} J(x) - R(x) & G(x) \\ -G^T(x) & 0 \end{pmatrix} \begin{pmatrix} \nabla H(x) \\ u \end{pmatrix} \quad (16)$$

where:

- $H(x) \geq 0$ is the internal Hamiltonian energy stored within each Clifford multivector neuron;
- $J(x) = -J^T(x)$ is a skew-symmetric interconnection matrix representing lossless energetic and rotational communication between neurons;
- $R(x) \succeq 0$ governs physical dissipation (conductance decay), ensuring strict Lyapunov passivity ($\dot{H} \leq y^T u$);
- (u, y) are effort-flow port variables enabling modular interconnectivity.

By replacing Euclidean linear projections with port-Hamiltonian coupling, inter-neuron communication will preserve bivector invariants and guarantee unconditional stability across arbitrary depth, unifying discrete associative scans with continuous physical dynamics.

7.3 Research Roadmap: BabyLM 100M, TinyStories, and ImageNet

To extend beyond the initial feasibility proofs established in this work, our active experimental agenda focuses on:

1. **Standardized Language Benchmarks:** Evaluating CSN against modern LLaMA-style architectures (RoPE + RMSNorm + SwiGLU) and Mamba SSMs across the **BabyLM Challenge (10M / 100M)** and **TinyStories** datasets on high-throughput NVIDIA RTX 4090 platforms.
2. **Spatial Scaling to ImageNet-1k:** Progressing from MOT16 pedestrian detection to full-scale hierarchical visual classification on ImageNet-1k to evaluate how dual-grade Clifford rotors scale against Vision Transformers (ViT) and modern ConvNets.
3. **Hardware-Native PHS Kernels:** Developing register-fused CUDA kernels for Port-Hamiltonian Dirac interconnections.

8 Conclusion

This paper has presented Version 1 of Clifford Space Networks (CSN), introducing continuous non-abelian Lie rotor scans as an analytically grounded, sub-quadratic alternative to unconstrained Euclidean networks.

By grounding state transitions in the rational Cayley transform on $\text{Spin}(n)$ and establishing strict algebraic isometry, CSN demonstrates remarkable sample efficiency and constant inference memory on compact feasibility benchmarks. Crucially, this work represents the foundational stage of a broader research trajectory. By explicitly identifying the inter-neuron communication bottleneck and proposing Port-Hamiltonian Dirac structures as the next evolutionary step, Clifford Space Networks offer a mathematically rigorous path toward analytical graph reasoning, physical system modeling, and scalable geometric intelligence.

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